A Robust Man-Made Target Detection Method Based on Relative Spectral Stationarity for High-Resolution SAR Images

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***Abstract*— The generality and robustness of a man-made target detection method are of high practical value in synthetic aperture radar (SAR) applications. However, current methods hardly adapt to different SAR images simultaneously because data property and target characteristics are not identical in images with various resolutions and scenes. The key to achieving high generality and robustness is extracting stable and invariant information in different SAR images. This article analyzes the scattering of man-made targets and natural backgrounds in SAR images and assumes that the noise characteristics are relatively stable and invariant when resolution and observing background change. Then, the relative spectral stationarity (RSS) is proposed based on 2-D spectrum analysis to measure the distance between the observed data and a manually generated noise. RSS takes the manually generated noise with known and definite properties as a standard, so it is irrelevant to image parameters and scenes. A low RSS indicates that the characteristics of observing data are dominated by noise, whereas a high value means that there may be targets in the observing data weakening the noise characteristics. An efficient segmentation algorithm Sparsity-based Format-free Segmentation within error *e* (SFFe) is proposed to process the RSS map and complete the detection method. SAR images with resolutions ranging from 0.1 to 8 m are employed as testing data. Various testing scenes are constructed to simulate different practical conditions. Experimental results validate that the proposed RSS-based method works well in SAR images with different resolutions, bands, and observing scenes, obtaining reliable and robust detection results and outperforming canonical methods on various criteria.**

***Index Terms*— Man-made targets' detection, noise generating, spectral stationarity, spectrum analysis, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR), known as the only way for all-time and all-weather Earth observation, is one of the most pivotal means of remote sensing [1]. One of the primary purposes of remote sensing using SAR images is to observe and evaluate human activities, where urban and ocean observation, field searching and rescuing, and damage evaluation are representative applications in both civilian and military aspects [2], [3]. Man-made targets, including buildings, facilities, and other unnatural objects, manufactured and deployed by humans, are the most significant signs of human activities. Hence, man-made target detection is of great research interest and value for SAR applications.

Three variables related to SAR image processing make robust man-made target detection difficult in practical applications. The first is resolution. There are more available SAR systems than before, and the available resolutions and bands are optional in a wide range [4], [5], [6]. Therefore, detection methods are required to adapt to commonly used resolutions. However, the data differences among various resolutions are far more significant than the resolutions themselves. The second is the target characteristic in SAR images. Because of the imaging mechanism, targets characteristic in SAR images are susceptible to poses of both the platform and targets themselves, such as target orientation and looking angels [7]. Moreover, prior information is insufficient in applications such as rescuing and damage evaluation, and targets of interest may differ from their original status due to damages [8]. This variable makes robustly describing SAR targets difficult. Finally, man-made targets can be deployed in any scene, so the surrounding environment and context of targets are variable. In summary, it is difficult to find invariant or stable information for detecting man-made targets robustly due to the optional resolutions of sensors, the pose sensitivity of targets, and various backgrounds in observing scenes [9].

Because of the three variables, it is still a challenging task for current research to achieve good overall performance on different SAR images, and the generality and robustness are limited. The key to high generality and robustness is to find stable and invariant information or indicators hidden beneath the SAR images of different resolutions and scenes. This article resorts to the spectral domain to look for universal and invariant information to describe differences between man-made targets and backgrounds, and formulates a reliable and robust method to separate them. In summary, the main work and highlighted contributions of this article are listed in the following.

1) First, this article analyzes the scattering of man-made targets and natural backgrounds in SAR images and

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illustrates that the noise characteristics are relatively stable and invariant when resolution, target class, platform pose, and background change.

2) Second, the relative spectral stationarity (RSS), the 2-D spectral similarity between the observing data and the generated noise, is proposed for man-made target detection.
3) Finally, an efficient and refined segmentation algorithm Sparsity-based Format-free Segmentation within error e (SFFe) is proposed to complete the detection method.
4) SAR images of different resolutions and scenes are selected, and six imaginary scenes simulating different practical situations are constructed. Performances of target detection and background filtering are tested and discussed comprehensively. Experimental results validate that the proposed RSS-based method achieves stable and robust performance when resolution, target scattering, and background change.

The remainder of this article is organized as follows. Section II lists the related works about robust man-made target detection. Section III discusses the characteristics of man-made targets and background, and formulates the RSS indicator. The detection framework based on RSS and SFFe is given in Section IV. In Section V, real SAR images of various bands and resolutions are employed to validate and illustrate the effectiveness and robustness of the proposed method. The conclusion is given in Section VI.

II. Related Works

Current methods for robust man-made target detection are roughly categorized into three groups, mathematical model-based, physical property-based, and machine learning-based.

The methods that directly deal with the original SAR data without additional transforms are categorized into mathematical model-based, and they model SAR data and clutter by mathematical distribution instead of directly describing targets of interest. Hence, such methods are not sensitive to target classes and are widely used in real applications for their generality and simplicity [1], [10]. However, the model's accuracy and adaptivity are challenged when the complexity of clutter and backgrounds change with resolutions and scenes. Refining schemes were proposed recently, such as super-pixel processing [11], additional censoring schemes [12], and other modified schemes [13]. However, the inherent drawbacks still limit the scope of the application. Thus, recent mathematical-based methods mainly focus on targets in monobackground, say maritime targets' detection [14], [15], [16]. Besides distribution models, matrix-based models are also employed to model the target and background. The robust principal component analysis (RPCA) methods model moving targets in SAR images as sparse matrices, whereas the stationary background is modeled as the low-rank matrix [17], [18]. RPCA-based methods take the background as invariant information when observing time changes, and corresponding algorithms, such as low-rank plus sparse decomposition, are employed to extract moving targets from stationary backgrounds [19], [20], [21]. However, RPCA-based methods need the support of time-series data to meet the low-rank-background requirement.

The methods that transform the original data to a certain form are defined as the physical property-based in this article. The transformed results are features or indicators that directly reveal the physical properties of targets and backgrounds. Such methods are commonly less sensitive to resolutions because physical properties maintain unchanged under different resolutions. Because of their robust and stable performance on common resolutions [22], [23], local spatial indicators, such as texture, are widely used to analyze local details and play a key role in applications such as damage evaluation [24], [25]. Recently developed regional spatial features, such as homogeneity and saliency [26], [27], are used for region comprehension and semantic analysis [28], [29], [30]. Generally, the local texture is more stable but less accurate, whereas the advanced regional feature is more meaningful and powerful under high-resolution conditions but more sensitive to image content and quality. In summary, because of the heavy speckle-noise [9], the pose sensitivity of man-made targets [7], and the complex scattering in urban regions, using spatial information only is hard to describe man-made target properties robustly.

Compared with spatial features, the complex spatial distribution hardly impacts the spectrum. Thus, spectral features are more stable and robust in SAR images of different resolutions and complex content [31], [32]. Time–frequency (TF) analysis, seen as the inverse process of aperture synthesizing by Laurent [33], acquires scattering information of targets under different azimuth angles from the decomposed subimages. The TF method reveals that the scattering of man-made targets varies with azimuth angles, whereas the natural background stays stationary. Azimuth stationarity successfully models man-made targets robustly, showing great potential for dealing with man-made targets in urban regions [34], [35].

The methods that can iteratively update themselves and optimize the results according to the extra or externally given information are defined as machine learning-based in this article. The sharply developed deep learning method is the most famous representative, such as convolutional neural networks [36]. Machine learning-based methods are famous for high precision and self-learning, and are widely used for isolated target detection [37] and building detection in complex backgrounds [38]. However, the performance and capability of machine learning-based methods highly rely on prior knowledge and training samples. Hence, they have the highest upper limit of performance when prior information and data are sufficient but a relatively low lower limit when samples are insufficient or unbalanced, resulting in risks in practical applications. The GAN-based balance learning method has been proposed to alleviate the reliance on data and balance the impact of sample distribution [39]; metalearning is developed to deal with general target detection tasks with few shots [40]. However, these refinements also need diverse and numerous training samples. Thus, they still fail to solve the mentioned issues simultaneously even though they can perfectly deal with one of them. This means that the current machine learning methods cannot fully take the conventional place due to the diversity of detection requirements and the complexity of SAR data [41].

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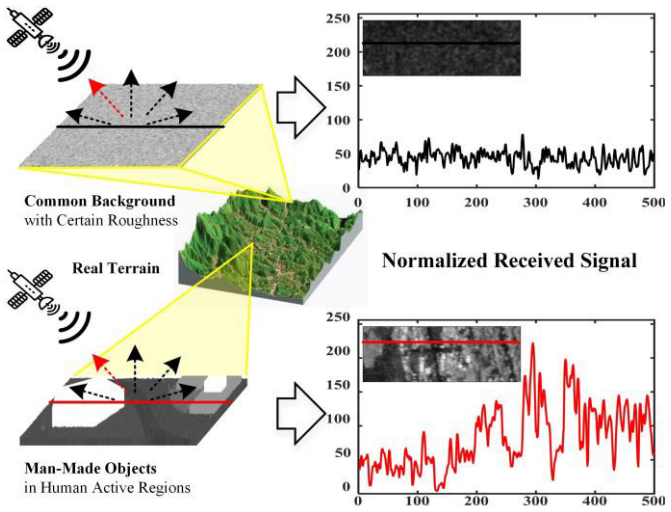


Fig. 1. Schematic of the receiving SAR signal of backgrounds with certain roughness and targets of interest in human-active regions.

Generally, current methods achieve great performance under certain situations, but the overall performance is still challenged when detection conditions change. Thus, the purpose of this article is to formulate a general man-made target detection method suitable for most currently used SAR images, and the overall performance is more emphasized in this article than the top performance.

III. Spectral Stationarity Relative to Noise

The characteristics of man-made targets, background, and noise in SAR images are analyzed in this section. Then, the spectral statistical characteristic of white noise is employed as the standard to formulate the similarity in the spectral domain between the observing data and noise. The RSS, relative to the white noise, is formulated based on this similarity.

A. Scattering Analysis and Noise Model

This article considers that the noise characteristics remain relatively consistent in different SAR images, as noise is not determined by specific factors but a combined result of the imaging mechanism, the radar system, and the observing objects. Ideally, the received SAR data merely reflect the noise information if the observing region is slightly textured. Hence, an indicator derived from the stable characteristics of noise can reveal the invariant and stable information hidden beneath various SAR images. Thus, this article takes noise as the key to solving the issues mentioned in the introduction rather than interference [17].

The differences between man-made targets and backgrounds are shown in Fig. 1. In this article, the background is defined relative to targets of interest and refers to regions of natural land covers with constant scattering power and certain roughness, such as water fields, bare land, and slightly vegetated areas.

The image information in background regions mainly reveals the noise information in detail besides the mean scattering power. In contrast, the detailed structures of man-made targets are recorded by the SAR signal, so target information is dominant in target regions, and the image details vary when the target categories and observing pose change. Then, the similarity between the observing data and noise property can be employed for robust target detection.

This article aims to formulate a method to robustly reveal and quantitively measure the subtle difference between the background and man-made targets. The product model in (1) is employed for mathematical analysis and validation [5], [42]

$$Y(i,j) = X(i,j) \cdot S(i,j) \tag{1}$$

where i and j represent the spatial position; Y stands for the observing amplitude or intensity [43]; X is the real scattering power of current observing objects; and S stands for the multiplicative speckle noise. According to Argenti et al. [44], the logarithmic process can transfer the multiplicative noise into the additive as shown in (2)

$$\begin{aligned}\log\left(Y_{ij}\right) &= \log\left(X_{ij} \cdot S_{ij}\right) = \log\left(X_{ij}\right) + \log\left(S_{ij}\right) \\ &= O_{ij} + N_{ij} = I_{ij}\end{aligned} \tag{2}$$

where the logarithmic observing value is denoted by O and the logarithmic noise is denoted by N.

Then, the logarithmic SAR image in this article is represented by (3), where $\mathbf{I}$ denotes the logarithmic intensity map derived from the original single-look complex data, and it is called intensity directly in the following text; $\mathbf{O}$ represents the real intensity produced by land covers, including targets of interest and backgrounds; $\boldsymbol{N}$ denotes the noise intensity; and H and W are the height and the width of the input image, respectively,

$$\begin{aligned}\mathbf{I} &= \left\{I_{ij} = O_{ij} + N_{ij} |\quad i = 1, 2, \ldots H, \quad j = 1, 2, \ldots W\right\} \\ &= \mathbf{O} + \boldsymbol{N}.\end{aligned} \tag{3}$$

B. Spectral Statistics of Targets and Noise

This article assumes that the additive logarithmic noise in (3) is white noise, which means that its spectrum distributes uniformly. Under this assumption, the power of logarithmic noise is represented by the standard deviation $\sigma_{\boldsymbol{N}}$. Then, it can be concluded that the noise power is equivalent to the input intensity power when land covers produce constant scattering intensity, as proven in the following equation:

$$\begin{aligned}\sigma_{\mathbf{I}} &= \sigma(\mathbf{I}) = \sigma(\mathbf{O} + \boldsymbol{N}) \\ &= \sqrt{\sum_{i=1}^{2d+1}\sum_{j=1}^{2d+1}\frac{\left(O_{ij} - E_{\mathbf{O}} + N_{ij} - E_{\boldsymbol{N}}\right)^2}{(2d+1)^2 - 1}} \\ &\overset{\mathbf{O}\equiv c}{=} \sqrt{\sum_{i=1}^{2d+1}\sum_{j=1}^{2d+1}\frac{\left(N_{ij} - E_{\boldsymbol{N}}\right)^2}{(2d+1)^2 - 1}} = \sigma(\boldsymbol{N}) = \sigma_{\boldsymbol{N}} \\ E_{\boldsymbol{N}} &= \frac{1}{(2d+1)^2}\sum_{i=1}^{2d+1}\sum_{j=1}^{2d+1}\left(N_{ij}\right) \\ E_{\mathbf{O}} &= \frac{1}{(2d+1)^2}\sum_{i=1}^{2d+1}\sum_{j=1}^{2d+1}\left(O_{ij}\right)\end{aligned} \tag{4}$$

where σ is the standard deviation operator; $\mathbf{I}$ denotes the input local intensity; and $\mathbf{O}$ and $\boldsymbol{N}$ denote the land cover scattering

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intensity and noise intensity, respectively. Standard deviation is implemented on a local scale, so a sliding window of size $(2d + 1)$ is used to sample data.

As mentioned in Section I, solely using spatial information is hard to model the data accurately and robustly, so this article resorts to the spectral domain to define noise characteristics. Fourier transformation shown in (5) is employed to acquire the spectrum

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{+\infty} f(t) e^{-\sqrt{-1}\omega t} d\omega. \tag{5}$$

The 2-D short-time Fourier transform (2-D STFT) is commonly adopted to process remote sensing images for balancing spectral and spatial resolutions. The spectral information of SAR images in local regions is derived by the following equation:

$$\begin{aligned} \mathbf{F_I} &= \mathcal{F}[\mathbf{I} - E(\mathbf{I})] = (F_{\mathbf{I}uv})_{(2d+1)\times(2d+1)} \\ F_{\mathbf{I}uv} &= F_{\mathbf{I}}(u, v) \\ &= \sum_{i=1}^{2d+1} \sum_{j=1}^{2d+1} \left(I_{ij} - E_{\mathbf{I}}\right) e^{-\sqrt{-1}\cdot 2\pi\left(\frac{ui}{2d+1} + \frac{vj}{2d+1}\right)} \end{aligned} \tag{6}$$

where $\mathcal{F}$ stands for discrete 2-D STFT; $\mathbf{I}$ is a local intensity matrix sampled from the original SAR image; $\mathbf{F}$ denotes the corresponding spectral matrix, and F is the element at a certain position in $\mathbf{F}$; and the subscripts u and v indicate the spectral coordinate. Considering the uniformity of symbol usage, $\sqrt{-1}$ is used as the imaginary unit instead of the commonly used j. The direct current (dc) component in the spectrum is removed by subtracting $E(\mathbf{I})$, and the bias of the mean value is eliminated.

According to (3), the input intensity image can be divided into the target and the background parts. The spectral characteristics of backgrounds can be approximately represented by noise because backgrounds are good noise carriers. Then, based on the linear property and additivity of Fourier transformation, (6) can be equally rewritten as follows:

$$\mathbf{F_I} = \mathbf{F_O} + \mathbf{F_N}. \tag{7}$$

The characteristics of $\mathbf{F_I}$ can be described in two dimensions. One is the dimension of value, reflecting the dispersion of spectral value; the other is the dimension of distribution, reflecting the uniformity of spectral energy distribution. In this article, the dimension of value is intuitively considered the vertical dimension, and the dimension of distribution is called the horizontal dimension.

Stationarity in the vertical dimension can be measured quantitatively by statistical variation, as shown in the following equation:

$$\sigma_{\mathbf{F}}^2 = \sigma^2(\mathbf{F_I}) = \sum_{u=1}^{2d+1} \sum_{v=1}^{2d+1} \frac{(F_{\mathbf{I}uv} - E_{\mathbf{F}})^2}{(2d+1)^2 - 1}. \tag{8}$$

Ideally, the intensity of natural land cover scattering, also referred to as clutter, is the summation of a constant scattering $\mathbf{O}$ and a noise intensity $\boldsymbol{N}$. The noise intensity can be divided into a constant $\bar{N}$ with a value equal to the mean of the noise and a zero-mean remainder $\boldsymbol{N}^*$. Constant value merely contributes to the dc component in the spectrum. Therefore, the vertical stationarity in background regions can be represented by the noise spectrum if the dc component F_{dc} is removed by (6), as shown in (9), and the detailed derivation is shown in the Appendix

$$\begin{aligned} \sigma_{\mathbf{F}}^2 &= \sigma^2(\mathbf{F_I}) = \sigma^2(\mathbf{F_O} + \mathbf{F}_{\bar{N}} + \mathbf{F}_{\boldsymbol{N}^*}) = \sigma^2(\mathbf{F}_{\boldsymbol{N}^*}) + \sigma^2(\mathbf{F}_{\mathrm{dc}}) \\ &= \sigma^2(\mathbf{F}_{\boldsymbol{N}^*}) - \frac{F_{\mathrm{dc}} \cdot E(\mathbf{F}_{\boldsymbol{N}^*})}{2d(d+1)} + \frac{F_{\mathrm{dc}}^2}{(2d+1)^2} \end{aligned} \tag{9}$$

where $\mathbf{F}_{\mathrm{dc}}$ stands for the merged spectrum containing only the dc component and F_{dc} denotes the dc component value.

The vertical stationarity is commonly lower in background parts than that in the target regions because the existence of targets complicates $\mathbf{F_O}$ and leads to a higher $\sigma_{\mathbf{F}}^2$. Besides, according to Parseval's theorem, spectral energy equals spatial energy. Though the dc component is removed by subtracting the mean value in the spatial domain, the spatial energy is unchanged. The spatial complexity synchronously increases the spectral variance. Hence, regions with high scattering power are more likely to produce high $\sigma_{\mathbf{F}}^2$. Thus, man-made targets that usually exhibit high scattering power in SAR images can be detected. From this view, spectral variance can serve as an indicator for target detection.

However, the background regions with high scattering power and complex texture also tend to produce a high $\sigma_{\mathbf{F}}^2$ value. In this case, the vertical stationarity fails to measure spectral stationarity accurately. Moreover, a nonzero $\mathbf{F_O}$ may decrease the variance in extremely rare situations. These two factors make the vertical stationarity less reliable than expected. Therefore, a stationarity measurement irrelevant to the spatial energy and complexity serves as a good supplement to improve the performance in complex regions.

Entropy describes stationarity by calculating the probability distribution in the horizontal dimension, and the spectral entropy is calculated as follows:

$$\mathrm{ENT}_{\mathbf{F}} = -\sum_{u=1}^{2d+1} \sum_{v=1}^{2d+1} \hat{F}_{\mathbf{I}uv} \log\left(\hat{F}_{\mathbf{I}uv} + \varepsilon\right) \tag{10}$$

where $\hat{F}$ stands for the normalized spectrum, and each value represents the percentage of the corresponding frequency component; ε is an adjustment term avoiding extreme values, and the value is set to 10^{-6} in the following experiments.

Compared with variance, entropy merely relates to the distribution and is independent of spatial energy. Hence, the entropy-based similarity is employed to modulate spectral variance for improving the capability of dealing with complex backgrounds.

C. Relative Spectral Stationarity

Noise in SAR images remains relatively stable and invariant when resolution, pose, and background change. Thus, this article takes the noise characteristics as a standard to measure relative entropy. In low-resolution SAR images, the intensity map can be divided into targets and noise or clutter by scattering power [45]. However, each pixel in high-resolution SAR images represents the summated power of targets or

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terrain surface and noise. Thus, it is difficult to separate noise accurately from the received data.

This article first assumes that the sampled local data from real SAR images consist of only noise. Therefore, the noise power σ_N equals $\sigma_{\mathbf{I}}$ and can be represented by the local image standard deviation. Then, white noise is generated manually based on the estimated σ_N and replaces the real sampled data. The standard noise with known characteristics is obtained in this manner, and the definite noise spectrum is calculated by the following equation:

$$\begin{aligned} \mathbf{F}_N &= \mathcal{F}(\boldsymbol{N}) = (F_{\boldsymbol{N}\mathrm{uv}})_{(2d+1)\times(2d+1)} \\ F_{\boldsymbol{N}\mathrm{uv}} &= F_N(u,v) = \sum_{i=1}^{2d+1}\sum_{j=1}^{2d+1} N_{ij} e^{-\sqrt{-1}\cdot 2\pi\left(\frac{ui}{2d+1}+\frac{vj}{2d+1}\right)} \end{aligned} \tag{11}$$

where $\boldsymbol{N}$ is the generated white noise data with the size of $(2d+1)$, the power is $\sigma_{\mathbf{I}}$, and the subscript N stands for noise. Based on (11), the noise entropy can be calculated by the following equation:

$$\mathrm{ENT}_N = -\sum_{u=1}^{2d+1}\sum_{v=1}^{2d+1} \hat{F}_{\boldsymbol{N}\mathrm{uv}} \log\left(\hat{F}_{\boldsymbol{N}\mathrm{uv}} + \varepsilon\right). \tag{12}$$

Based on the computable entropy, the similarity between noise and real data can be measured quantitively. An exponential form is adopted in this article, as shown in the following equation:

$$\mathrm{Sim} = \frac{e^{\mathrm{ENT}_{\mathbf{F}}}}{e^{\mathrm{ENT}_N}} = e^{\mathrm{ENT}_{\mathbf{F}}-\mathrm{ENT}_N} = \exp\left(\mathrm{ENT}_{\mathbf{F}} - \mathrm{ENT}_N\right) \tag{13}$$

where $\mathrm{ENT}_{\mathbf{F}}$ represents the entropy of real data, ENT_N stands for the entropy of the generated noise, and Sim represents the exponential similarity. If Sim is high, the hypothesis of the observing data contains only noise is true, so the local region is more closed to white noise. Otherwise, the hypothesis is not tenable. Mathematically, Sim represents the relative entropy over the generated noise. Compared with the absolute entropy value, Sim is adjusted to a reasonable and stable range. Hence, the spectral variance is modulated more robustly and predictably.

The reciprocal of Sim is employed as a modulation to balance the increasing spectral variance caused by the high background scattering power. Then, the RSS is defined as follows:

$$\begin{aligned} &\mathrm{RSS} \\ &= \begin{cases} 0, & \mathrm{ENT}_{\mathbf{F}} > R \\ \sigma_{\mathbf{F}}^2 \cdot \mathrm{Sim}^{-1} = \sigma_{\mathbf{F}}^2 \cdot \exp\left(\mathrm{ENT}_N - \mathrm{ENT}_{\mathbf{F}}\right), & \mathrm{ENT}_{\mathbf{F}} \le R \end{cases} \\ &\mathbf{S} \\ &= \left\{\mathrm{RSS}_{xy} | x = 1, 2, \ldots H, y = 1, 2, \ldots, W\right\} \end{aligned} \tag{14}$$

where R is the truncation parameter and $\mathbf{S}$ denotes the RSS map.

The modulating term Sim^{-1} is related to the exponential distance of entropy and the observing scale d. Fig. 2(a) shows the modulating value surface of entropy difference and d.

According to related research on the relationship between observing scale and spectral information exploitation [31], a moderate d is recommended, and the modulating curve with d equaling 5 is shown in Fig. 2(b).

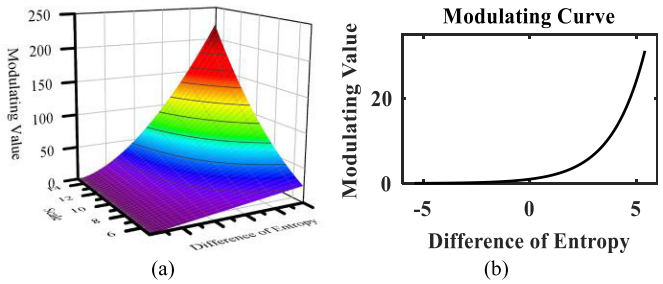


Fig. 2. Modulating value surface and curve of entropy-similarity-based modulation. (a) Modulating value surface. (b) Modulating curve ($d = 5$).

The entropy function has an upper limit, and only uniformly distributed data can reach this limit. Commonly, the spectrum entropy of the generated noise is greater than that of real data. Considering unpredictable situations, a truncation parameter R is introduced to set regions with exceptionally high entropy values to noise directly. The definition is shown in the following equation:

$$R = K \cdot \max(\mathrm{ENT}) = 2 \times K \times \log(2d+1) \tag{15}$$

where K ranges from 0 to 1 and serves as a truncation rate to determine R. The truncation parameter filters the regions with a spectral entropy higher than R because such a high entropy is close to white noise. K can be seen as a normalized threshold, and it is manually determined.

Physically, the proposed RSS is a 2-D statistical similarity in the spectral domain between local data and white noise. For an observed region, a low RSS illustrates that the local data mainly exhibits noise characteristics. Thus, noise is dominant in this region. The noise characteristics are weakened if man-made targets locate in this region, leading to a high RSS. RSS takes the manually generated noise with known and definite properties as a standard, so it is irrelative to imagining parameters, poses, and backgrounds. Therefore, RSS can be used as a robust measurement for man-made target detection.

IV. Man-Made Targets' Detection Based on RSS

Based on the RSS, this article proposes a corresponding method to detect man-made targets in SAR images of various resolutions robustly. This method extracts candidate targets from SAR images and filters' backgrounds simultaneously for following processing and applications.

The detection method consists of four steps, and the flowchart is given in Fig. 3. The first step is preprocessing, including geometry and radiational correction, and logarithmic filtering. It is worth noting that noise filtering is not a necessary step due to the mechanism of RSS. Thus, this noise-filter-free method is suitable for remote sensing image processing from the perspective of efficiency.

The second step is RSS map extraction, and the detailed procedure is given in the lower half of Fig. 3. There are two branches in this step after data sampling using a sliding window with the size of $(2d+1)$: the image branch and

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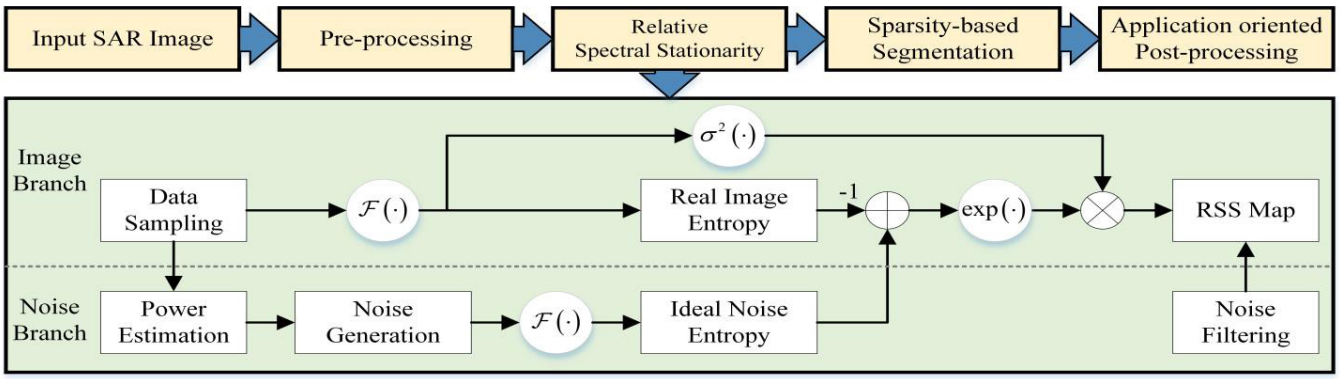


Fig. 3. Flowchart of the man-made detection framework based on RSS.

the noise branch. In the image branch, Fourier transformation is employed to obtain the spectrum from real data and calculate the corresponding statistical measurements. In the noise branch, random value generation is employed to generate white noise based on estimating the noise power from real data. Then, the spectral entropy of noise is calculated to get the final RSS map.

The RSS map shows the spectral distance to noise, and a high value indicates that the corresponding pixel is more likely to belong to a man-made target. Hence, a segmentation algorithm is employed in the following step to select candidate man-made targets and filter background parts. Because RSS is generated based on the spectrum, the spatial information is so insufficient that histogram-based segmentation methods are more suitable for processing the RSS map than the region-based. However, the RSS data are noninteger quantized and unevenly distributed in an unfixed range. Thus, it is difficult for conventional histogram-based methods to acquire an accurate histogram without time-consuming sorting. This article refines the histogram-based segmentation method and proposes the SFFe. Compared with conventional histogram-based segmentation, SFFe searches the threshold from the input data itself instead of attaining the entire histogram, avoiding the time-consumed sorting stage [46] and the accuracy loss caused by histogram estimation. The algorithm is shown in the following.

Compared with conventional histogram-based methods, the SFFe is a fast implementation scheme that trades adaptivity for efficiency and compatibility by a manually determined p, and the error is well controlled within e. Though the adaptivity is limited by p, the choice of p has a high tolerance. Namely, the p in a wide range can acquire similar results. That is because the data in SAR images distribute so dispersedly that there is a wide gap between target and background distributions. If the gap is far greater than error tolerance e, most p located in this gap will get the same result. Meanwhile, a minor change of p may filter some isolated points belonging to targets, but the feature points with high power are still reserved. Therefore, these targets are still detectable. This illustrates that SFFe has a good tolerance for spatial information due to the wide selection range of p.

The final stage is application-oriented postprocessing, such as size selection, structure refinement, and target classification.

Algorithm 1 Sparsity-Based Format-Free Segmentation Within Error e

1. **Input:** Serialized image vector $\mathbf{S}=\{S_t|t = 1,2,\ldots, H \times W\}$, error tolerance e, estimated sparsity rate p;
2. **Initialization:** $T_{\min} = \min(\mathbf{S})$, $T_{\max} = \max(\mathbf{S})$, $T^* = (T_{\max} + T_{\min})/2$, irritation index $t = 1$

3. **BEGIN:**
4. Let $T = S_t$: **If:** $T < T_{\max}$ and $T > T_{\min}$: **continue** to **5**;
   Else: $t = t+1$, **repeat 4**;
5. **Sparsity Estimation:** Calculate Current Sparsity CS by (16):

$$\mathbf{R} = \left\{ R_i | R_i = \begin{cases} 0, & S_i < T \\ 1, & S_i \geq T \end{cases}, i = 1, 2, \ldots, H \times W \right\}, \quad CS = \frac{\sum_{i=1}^{H \times W} R_i}{H \times W}. \tag{16}$$

6. **Error Check: If:** $|p\text{-}CS| > e$: update T by (17):

$$T = \frac{T_{\max} + T_{\min}}{2}, T_{\max} = \begin{cases} T_{\max}, & p < CS \\ T, & p > CS \end{cases}, T_{\min} = \begin{cases} T, & p < CS \\ T_{\min}, & p > CS \end{cases}. \tag{17}$$

   Else: $T^* = T$, **go to 8-END**;
7. **IF:** $t < H \times W$: $t = t+1$, **return** to **4**;
8. **END.**

Output: T^*.

The specific processing depends on the requirements of a definite task. Details of such postprocessing are not necessary and important in this article. Hence, postprocessing algorithms for specific tasks are directly employed without discussion in this article to fulfill the whole processing flow and display the method performance in real applications.

V. Experiments and Discussion

In this section, real SAR images of various bands and resolutions are employed as the testing data, and different testing scenarios are constructed based on the samples of man-made targets and backgrounds. Canonical indicators are selected as comparisons to evaluate the proposed RSS-based method comprehensively. The best performance, average effectiveness, robustness, and practical potential are tested and discussed in this part.

A. Target Detection Test Using HR SAR Images

A meter-level L-band UAVSAR image released by the Jet Propulsion Laboratory (JPL) is employed to test the performance of the proposed RSS. The observing scene is Half-moon Bay in California, covering multiclass targets and

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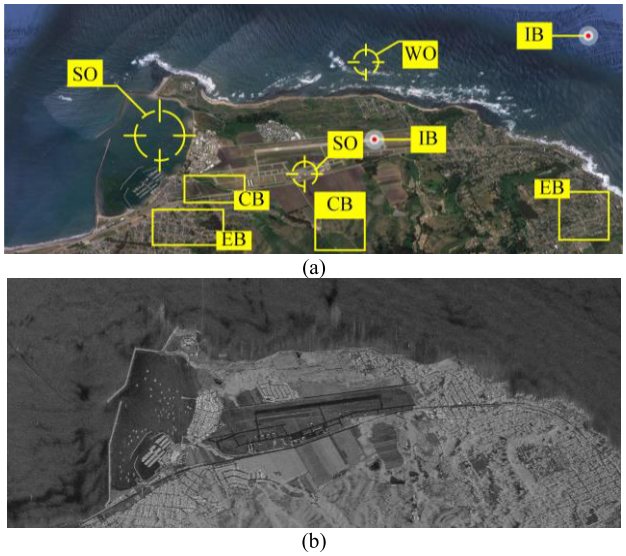


Fig. 4. Meter-level testing region. (a) Optical image with area indication. (b) Logarithmic SAR intensity image.

different background types. The optical image of this area and the logarithmic SAR intensity image that serves as the input data are shown in Fig. 4.

For a thorough and comprehensive comparison to canonical methods, this article divides man-made targets and backgrounds into several classes to construct detection situations with different difficulties. The samples are marked in Fig. 4(a). According to the local contrast and detection difficulty, man-made targets are divided into strong targets (STs) and weak targets (WTs). The strong targets noted by ST have strong and obvious scattering power in SAR images, whereas the weak targets noted by WT exhibit low power and ambiguous texture. IB denotes the ideal background with low power and smooth texture, commonly including water regions, roads, and so on. Most man-made targets located in ideal backgrounds are easier to detect than in other backgrounds. CB means complex background parts. Pixels in these parts share similar scattering power with man-made targets, and the texture is more complex than an ideal background. Slightly vegetated regions and simple compound regions belong to this class. EB means extremely complex backgrounds, such as urban regions and heavily vegetated regions. Man-made targets and backgrounds are mixed in this area, bringing difficulties to detection tasks.

Samples of strong and weak targets extracted from Fig. 4(b) are shown in Fig. 5, and samples of three backgrounds are shown in Fig. 6. The RSS map of the UAVSAR is shown in Fig. 7, and the corresponding RSS results of these samples are shown in Figs. 8 and 9.

The result illustrates that the RSS map enhances the man-made targets with either strong or weak scattering power and suppresses backgrounds of different complexities. However, the dynamic range of the RSS map is so wide that it is difficult to evaluate the performance visually. Thus, a set of indices are employed for thorough evaluation and comparison on the statistical level.

Six representative scenes of SAR target detection are constructed based on the two types of targets and the three types of backgrounds in Fig. 4(b). The target samples from Fig. 5 and the background samples from Fig. 6 are composited pairwise to generate the six imaginary scenes. All these samples serve as the ground truth to evaluate the performances in the six virtual scenes, e.g., the RSS distribution curves extracted

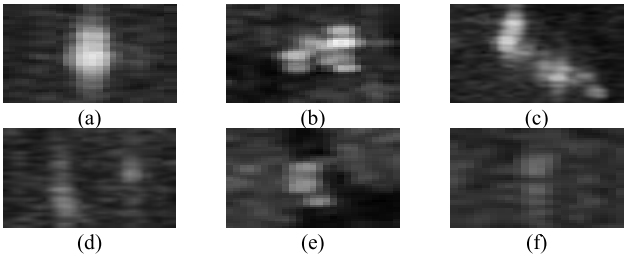


Fig. 5. Intensity samples of man-made target patch. (a)–(c) Targets with strong scattering power. (d)–(f) Targets with weak scattering power.

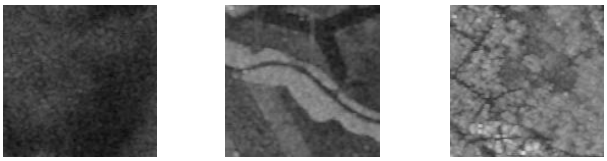


Fig. 6. Intensity samples of background patch. (a) Ideal background: water regions and roads. (b) Complex background: vegetated and compounded regions. (c) Extremely complex background: urban regions.



Fig. 7. RSS map of the meter-level UAVSAR image.

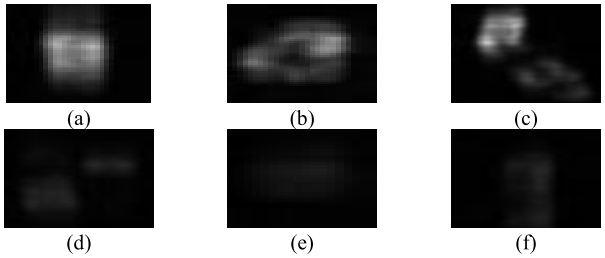


Fig. 8. RSS samples of man-made target patch. (a)–(c) Targets with strong scattering power. (d)–(f) Targets with weak scattering power.

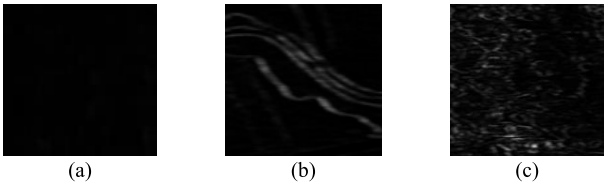


Fig. 9. RSS samples of background patch. (a) Ideal background. (b) Complex background. (c) Extremely complex background.

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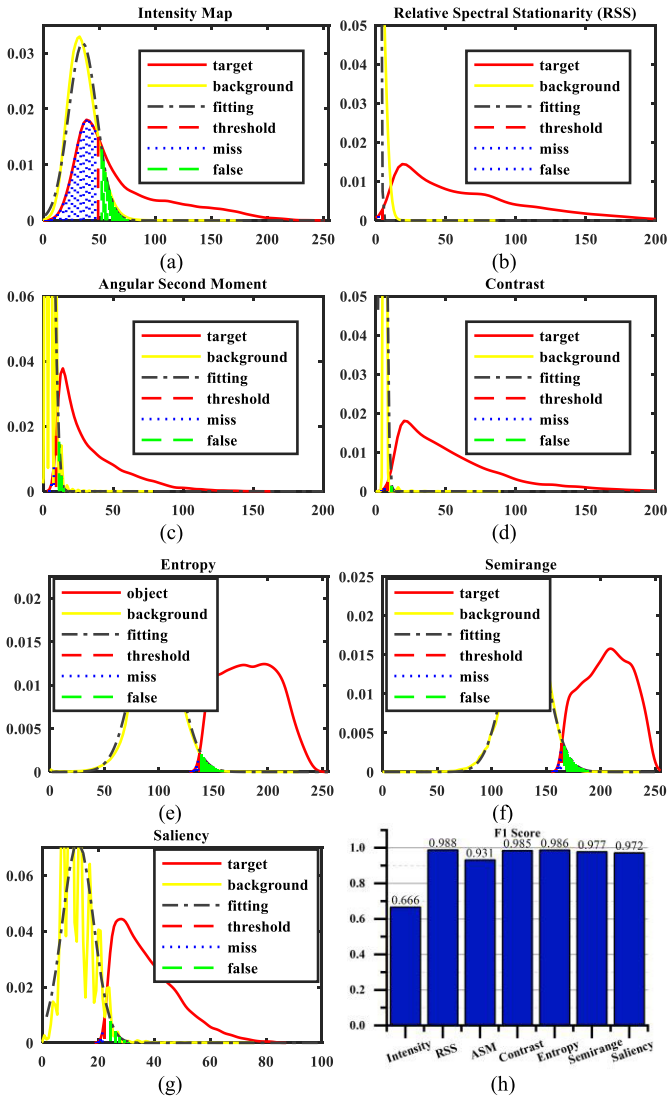


Fig. 10. Statistical results of RSS and comparisons on the scene of strong targets in ideal background. (a) Intensity map. (b) RSS. (c) GLCM ASM. (d) GLCM CON. (e) GLCM ENT. (f) Semivariogram semirange. (g) Spectral saliency. (h) F_1 score.

from Figs. 8 and 9 are pairwise combined to evaluate the performance on the statistical level. The canonical and widely used indicators in both the spatial domain and the spectral domain for target detection are employed as the comparisons, including GLCM-based statistics, semivariogram-based indicators, and spectrum-based saliency. According to the research conclusions of these indicators and practical experience, the angular second moment (ASM), contrast (CON), entropy (ENT) of GLCM, and the semirange of semivariogram are selected. Then, the proposed RSS and the comparisons are tested on the six scenes.

1) Strong Targets in Ideal Background: The first scene is strong targets in the ideal background, simulating an ideal detection scene in SAR images, such as ship detection in a low sea state. The statistical results of RSS and other comparisons are shown in Fig. 10.

Fig. 10(a) gives the probability density function (pdf) curve of strong target pixels and ideal background pixels in the logarithmic SAR intensity image, showing the data distribution of both targets and backgrounds. It should be noticed that all data are linearly corrected to the image domain [0, 255] for the simplicity of display, and the correction introduces no influence on the following evaluation because the linear transformation does not change the results of probability calculation. In Fig. 10(a), the pdf_O of targets is denoted by the red solid line, whereas the background pdf_B is denoted by the yellow solid line. For the convenience of calculation, a Gaussian curve represented by the black dashed-dotted line is used to fitting the background pdf_B. It should be noticed that the Gaussian curve fits the background well, validating that the logarithmic intensity of background and noise obey Gaussian distribution in the spatial domain. The intersection of the pdfs of targets and background indicated by the red dashed line is considered as the optimal detection threshold, T^*. Then, the area filled with blue dotted lines represents the missing detection rate (MR), whereas the green-dashed-line-filled part shows the false detection rate (FR) under this optimal threshold. The calculation equations are given in the following equation:

$$\mathrm{MR} = \int_{-\infty}^{T^*} \mathrm{Pdf}_O(x)\,dx, \quad \mathrm{FR} = \int_{T^*}^{+\infty} \mathrm{Pdf}_B(x)\,dx. \tag{18}$$

Fig. 10(b) shows the RSS result, and Fig. 10(c)–(g) shows the results of comparisons. Intuitively and qualitatively, both RSS and comparative indicators show the excellent capability of strong target detection in ideal backgrounds, and this conclusion is accordant with common knowledge.

The F_1 score in Fig. 10(h) gives a quantitative evaluation for these indicators, and F_1 is calculated by the following equation:

$$F_1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{19}$$

where recall and precision are calculated based on (20)–(22) [47]

$$\mathrm{TP} = \int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx, \ \mathrm{FN} = \int_{-\infty}^{T^*} \mathrm{Pdf}_O(x)\,dx = \mathrm{MR}$$
$$\mathrm{TN} = \int_{-\infty}^{T^*} \mathrm{Pdf}_B(x)\,dx, \ \mathrm{FP} = \int_{T^*}^{+\infty} \mathrm{Pdf}_B(x)\,dx = \mathrm{FR} \tag{20}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} = \frac{\int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx}{\int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx + \int_{-\infty}^{T^*} \mathrm{Pdf}_O(x)\,dx} = \int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx \tag{21}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} = \frac{\int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx}{\int_{T^*}^{+\infty} \mathrm{Pdf}_O(x)\,dx + \int_{T^*}^{+\infty} \mathrm{Pdf}_B(x)\,dx}. \tag{22}$$

Fig. 10(h) shows that the proposed RSS shares a similar performance under the F_1 criterion with comparative indicators. This result illustrates that RSS can replace other indicators in the scene of strong targets in the ideal background and achieve a minor increase in F_1 score.

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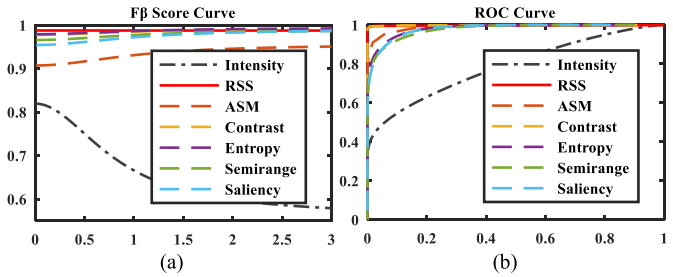


Fig. 11. Evaluation results of each indicator in the first testing scene. (a) F_β score. (b) ROC curve.

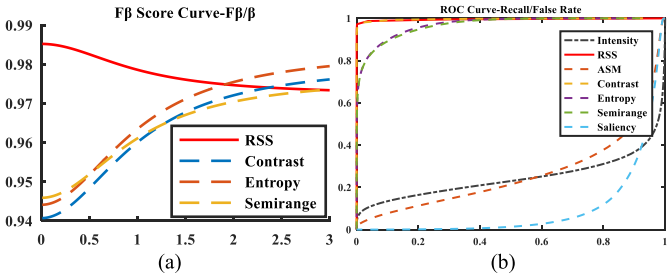


Fig. 12. Evaluation results of each indicator in the second testing scene. (a) F_β score. (b) ROC curve.

Fig. 11 gives a more comprehensive evaluation using the F_β score and receiver operating characteristic (ROC) curve. The F_β score introduces the weight β that represents that recall is β times as important as precision, as shown in (23). When β equals one, F_β turns to F_1

$$F_\beta = \left(1+\beta^2\right) \times \frac{\text{Precision} \times \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}. \tag{23}$$

F_β with β from 0 to 3 is shown in Fig. 11(a), illustrating that RSS is superior to others when $\beta < 1.5$ and is surpassed by ENT and CON when β is greater, though the differences among these indicators are very slight except for ASM. The ROC curve in Fig. 11(b) shows the relationship between recall and false alarm under different thresholds T. The result illustrates that the RSS and CON will reach a high recall value when the false alarm increases.

In general, Figs. 10 and 11 show the upper performance limit of target detection using different indicators. The results validate that either RSS or the comparisons perform well under ideal conditions. Even the intensity can detect targets in ideal backgrounds because its ROC curve is convex. Quantitatively, RSS outperforms others a little under most criteria.

2) Strong Targets in Complex Background: The second scene is strong targets in complex background, simulating a common detection scene like man-made targets in natural background. In this scenario, the distribution of strong target data remains unchanged, whereas the background complexity increases. In this scenario, the pixel values of targets and background in intensity, GLCM ASM, and saliency are mixed severely. It is impossible to define an optimal threshold, so pdfs and F score curves of these four indicators are omitted, but the threshold-free ROC curve is remained to illustrate their performances, as shown in Fig. 12.

Intuitively, the performances of all these indicators degrade to some extent due to the increased background complexity and detection difficulty. The quantitative F_β score in Fig. 12(a) shows that RSS remains its good performance when β is small, and the score is always greater than 0.97, where the decrease is less than 1% compared with that in Scene 1. The ROC curve in Fig. 12(b) divided these indicators into three groups. RSS and GLCM CON still perform the best and reach a good recall value with a small false alarm, whereas ENT and semirange reach their top recall values when the false alarm is greater than 40%. The ROC curves of intensity, ASM, and saliency are concave, illustrating that they are not suitable for target detection in this scene for the false alarm, which is always greater than recall.

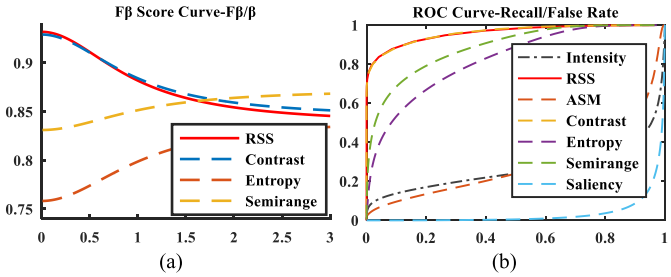


Fig. 13. Evaluation results of each indicator in the third testing scene. (a) F_β score. (b) ROC curve.

In general, Fig. 12 displays the average performance of RSS and other canonical indicators for SAR target detection. The results validate that RSS and CON own the best detection capability, where RSS has a better performance on balancing recall and precision. Quantitatively, RSS still outperforms others a little under most criteria. Saliency, a powerful local contrast indicator, scores a low value according to Fig. 12. The reason is that all the indicators are performed on the whole image, but only the most salient part of the image is extracted by saliency.

3) Strong Targets in Extremely Complex Background: The third scene is strong targets in an extremely complex background, simulating man-made targets in an urban environment. This scene is important and common in SAR image applications. The statistical results of this scenario are given in Fig. 13.

In this scenario, the background complexity increases greatly, and the data distribution is more complex than in the former two scenes, bringing much more difficulty to targets' detection. The F_β score in Fig. 13(a) validates this phenomenon quantitatively. The F_β scores of RSS and CON drop to 0.9 when β is greater than 0.8 and continue decreasing with the increasing β. RSS, CON, and semirange share a similar F_β score when β reaches 1.5. Semirange wins a little when β is greater. From the view of F_β, RSS and CON take the superior place most of the time. The ROC curve in Fig. 13(b) shows a similar order of these indicators to Fig. 12, but the performances are decreased. Semirange performs better than ENT in extremely complex backgrounds but still worse than RSS and CON.

In general, RSS, CON, ENT, and semirange maintain their capability of targets' detection in extremely complex

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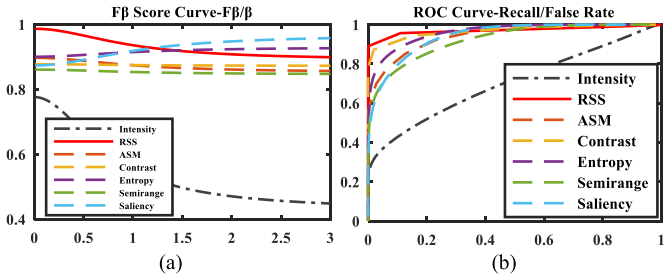


Fig. 14. Evaluation results of each indicator in the fourth testing scene. (a) F_β score. (b) ROC curve.

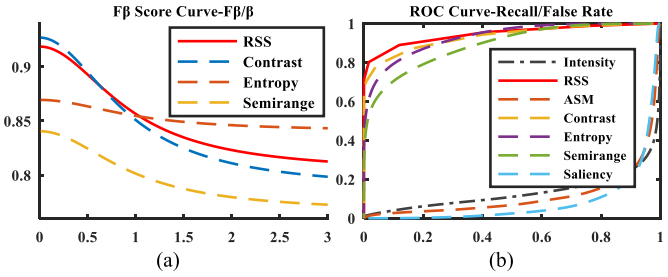


Fig. 15. Evaluation results of each indicator in the fifth testing scene. (a) F_β score. (b) ROC curve.

backgrounds because the distinct patterns and high scattering of strong targets produce high responses in these indicators. RSS and CON perform similarly and outperform other indicators in this scenario, validating their powerful capability for targets' detection using meter-level SAR images.

According to the experiments in these three scenarios, strong targets in ideal, complex, and extremely complex backgrounds, RSS shows its effective and robust performance, especially under the F_β criterion. RSS also performs best under the ROC criterion though it is followed tightly by CON. All these three experiments validate RSS as an excellent global indicator for strong targets' detection using SAR images.

4) Weak Targets in Ideal Background: For weak targets' detection, another three scenes are constructed. The fourth scene is weak targets in the ideal background. In this case, the pdf curve of targets changes, and the background curve maintains. The structure and texture of targets are blurred due to the low scattering power, so the performances of RSS and the comparisons deteriorate, and the F_β score and the ROC curve in Fig. 14 reflect this degradation. According to Fig. 14(a), RSS still takes the superior place in F_β when β is smaller than 1.2. The CON performance is not as good as before, whereas ENT and saliency perform well under a big β. The ROC curve in Fig. 14(b) shows that the recall of RSS reaches an accepted value sharply when the corresponding false alarm rate increases a little, and RSS outperforms all others when the false alarm rate is less than 30. RSS is surpassed by saliency and ENT when the tolerance of false alarms is more than 30%.

Generally, though the detection difficulty increases due to the weak scattering power and vague structure of targets, all these indicators still maintain their capability to a great extent because of the ideality of the background. RSS shows excellent performance and robustness and outperforms all others in this scenario.

5) Weak Targets in Complex Background: The fifth scene is weak targets in complex background, and the results are given in Fig. 15.

There is a sharp performance decrease in all indicators in this scenario because the detection difficulty increases further. The best detection rate is lower than 85% and the average is lower than 80% for the first time. ASM and saliency are not suitable for targets' detection in this scenario. According to Fig. 15(a), RSS shows a similar performance to CON, whereas ENT shows a more stable performance under the F_β criterion.

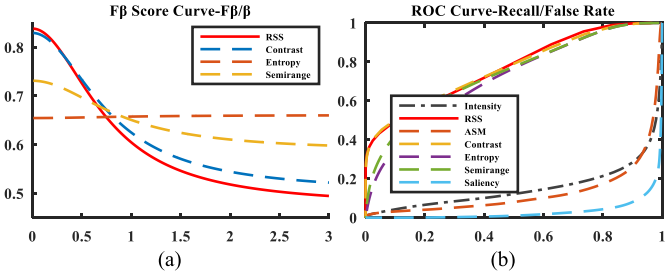


Fig. 16. Evaluation results of each indicator in the sixth testing scene. (a) F_β score. (b) ROC curve.

According to the ROC curve in Fig. 15(b), RSS is still superior to others and obtains the highest recall when the false alarm rate tolerance is lower than 40%.

6) Weak Targets in Extremely Complex Background: The last scene is weak targets in an extremely complex background, which is the most difficult scenario for target detection using SAR images. Most pixels of targets are submerged by the high scattering power and complex texture pattern of background targets, and the results in Fig. 16 support this. In this scene, the pdf curves of targets and background overlay seriously, and the optimal threshold T^* fails to separate them with a good performance. To some extent, weak targets are hardly detected in extremely complex backgrounds. However, these indicators still can be compared quantitatively by statistics.

The F_β score and the ROC curve are given in Fig. 16(a) and (b), respectively. The F_β score curve shows that semirange maintains the performance when the background complexity increases, and ENT is the most stable indicator when β changes. RSS reaches the highest F_β score with an ideal β value but is surpassed a little by CON when β increases. RSS still takes the leading place under the ROC criterion, followed tightly by CON. Both RSS and CON reach 80% recall when the false alarm rate is 60%, showing the difficulty of target detection in this scene. ENT under the ROC criterion performs not as well as that under the F_β score.

Other specific statistics and evaluation parameters are employed and formulated to make the evaluation more comprehensive and intuitive. The overall experimental results of these seven indicators in the six testing scenes are given in Table I, and the optimal and the second optimal indicator under a certain criterion are marked in black and gray, respectively.

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TABLE I

STATISTICAL RESULTS OF ALL INDICATORS IN SIX TESTING SCENES

Testing Scene		Index	Intensity	RSS	GLCM ASM	Contrast	Entropy	Semirange	Spectral Saliency
Strong Objects	Ideal Background	Missing Rate	43.93%	1.24%	4.41%	0.75%	0.63%	1.10%	0.96%
		False Rate	12.36%	1.23%	9.78%	2.29%	2.12%	3.51%	4.74%
		F_1 Score	0.6657	0.9876	0.9309	0.9849	0.9863	0.9772	0.9720
		$F_{0.5}$ Score	0.7501	0.9877	0.9165	0.9804	0.9820	0.9703	0.9613
		F_2 Score	0.5984	0.9875	0.9457	0.9894	0.9907	0.9842	0.9829
		$F_{Mean(\beta\in(0,2))}$	0.6531	0.9876	0.9353	0.9861	0.9875	0.9792	0.9752
		$F_{Stability(\beta\in(0,2))}$	5.1606	19.0319	8.5197	10.8706	10.9271	10.0038	9.1214
		Leading Period	0	59%	0	0	41%	0	0
		$Recall_{FalseRate=1\%}$	39.51%	98.71%	85.15%	99.25%	78.40%	76.19%	73.34%
		$FalseRate_{Recall=98\%}$	93.20%	<0.01%	15.64%	0.01%	25.06%	30.61%	19.05%
	Complex Background	Missing Rate		2.79%		1.97%	1.63%	2.32%	
		False Rate		1.46%		6.18%	5.83%	5.59%	
		F_1 Score		0.9786		0.9600	0.9634	0.9610	
		$F_{0.5}$ Score		0.9825		0.9483	0.9517	0.9519	
		F_2 Score		0.9746		0.9720	0.9754	0.9704	
		$F_{Mean(\beta\in(0,2))}$		0.9775		0.9635	0.9669	0.9638	
		$F_{Stability(\beta\in(0,2))}$		11.1316		8.9317	8.9330	9.4258	
		Leading Period	0	93%	2.5%	0	4.5%	0	0
		$Recall_{FalseRate=1\%}$	7.48%	97.21%	2.58%	95.81%	70.69%	70.89%	<0.01%
		$FalseRate_{Recall=98\%}$	99.95%	0.11%	98.72%	0.17%	19.74%	20.70%	98.81%
	Extremely Complex Background	Missing Rate		16.30%		15.65%	15.63%	12.72%	
		False Rate		6.11%		6.43%	26.91%	17.74%	
		F_1 Score		0.8819		0.8842	0.7986	0.8514	
		$F_{0.5}$ Score		0.9112		0.9106	0.7738	0.8391	
		F_2 Score		0.8543		0.8593	0.8250	0.8641	
		$F_{Mean(\beta\in(0,2))}$		0.8747		0.8777	0.8067	0.8551	
		$F_{Stability(\beta\in(0,2))}$		7.1683		7.3737	7.3982	8.8288	
		Leading Period	0	29%	0	60%	0	11%	0
		$Recall_{FalseRate=1\%}$	7.48%	75.85%	3.56%	76.09%	17.94%	34.22%	<0.01%
		$FalseRate_{Recall=98\%}$	99.85%	39.64%	98.63%	29.81%	62.95%	50.49%	99.90%
Weak Objects	Ideal Background	Missing Rate	57.17%	10.99%	14.77%	12.79%	7.04%	15.39%	3.19%
		False Rate	12.36%	1.23%	9.78%	12.2%	10.31%	13.70%	14.04%
		F_1 Score	0.5519	0.9357	0.8741	0.8746	0.9146	0.8533	0.9183
		$F_{0.5}$ Score	0.6676	0.9655	0.8877	0.8762	0.9059	0.8577	0.8908
		F_2 Score	0.4704	0.9078	0.8608	0.8731	0.9235	0.8489	0.9475
		$F_{Mean(\beta\in(0,2))}$	0.5420	0.9285	0.8705	0.8742	0.9172	0.8520	0.9272
		$F_{Stability(\beta\in(0,2))}$	4.5794	7.1404	8.6719	12.9709	9.5312	10.9101	7.1930
		Leading Period	0	60%	0	0	0	0	40%
		$Recall_{FalseRate=1\%}$	28.21%	89.00%	56.87%	78.64%	68.08%	55.20%	51.10%
		$FalseRate_{Recall=98\%}$	92.80%	11.05%	36.97%	49.98%	24.190%	40.79%	32.15%
	Complex Background	Missing Rate		19.77%		21.36%	15.95%	23.42%	
		False Rate		7.09%		6.18%	12.62%	14.53%	
		F_1 Score		0.8566		0.8509	0.8547	0.814	
		$F_{0.5}$ Score		0.8929		0.8950	0.8635	0.8244	
		F_2 Score		0.8231		0.8110	0.8461	0.7796	
		$F_{Mean(\beta\in(0,2))}$		0.8481		0.8411	0.8524	0.7957	
		$F_{Stability(\beta\in(0,2))}$		6.7590		6.3801	9.5455	7.6483	
		Leading Period	0	21%	0	31%	48%	0	0
		$Recall_{FalseRate=1\%}$	1.40%	71.48%	0.78%	70.17%	55.78%	47.95%	<0.01%
		$FalseRate_{Recall=98\%}$	99.99%	41.19%	99.43%	68.20%	40.08%	52.51%	99.09%
	Extremely Complex Background	Missing Rate		52.69%		49.86%	33.92%	41.36%	
		False Rate		9.11%		10.30%	34.86%	21.54%	
		F_1 Score		0.6048		0.6250	0.6577	0.6509	
		$F_{0.5}$ Score		0.7263		0.7335	0.6558	0.6969	
		F_2 Score		0.5182		0.5444	0.6595	0.6105	
		$F_{Mean(\beta\in(0,2))}$		0.5937		0.6129	0.6582	0.6415	
		$F_{Stability(\beta\in(0,2))}$		4.4785		4.6895	12.6748	6.3190	
		Leading Period	0	16%	0	21%	54.50%	8.50%	0
		$Recall_{FalseRate=1\%}$	1.40%	33.27%	1.36%	32.62%	7.82%	12.18%	<0.01%
		$FalseRate_{Recall=98\%}$	99.95%	73.50%	99.40%	83.92%	78.76%	77.960%	99.94%
Number of Optimal Items (*OI*)			0	29	0	11	15	3	2
Number of Second Optimal Items (*SOI*)			0	15	1	15	16	10	3
Number of Invalid Scenes (*IS*)			4	0	4	0	0	0	4
Custom Score:			0	21.9	0.1	11.1	13.8	4.8	0.7

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If an indicator is unsuitable for detection in a testing scene, the corresponding blank is marked by a slash.

The missing rate and the false rate, calculated by (18), are employed as the direct evaluation parameters. Based on F_β in (23), three commonly used specific β values are selected: F_1 score considers recall and precision equally, $F_{0.5}$ focuses on precision, and F_2 focuses more on recall than precision. $F_{\text{Mean}(\beta\in(0,2))}$ is formulated to show the average score in F_β, whereas $F_{\text{Stability}(\beta\in(0,2))}$ shows the stability in F_β with the range of β (0, 2), as shown in the following equation:

$$F_{\text{Mean}(\beta\in(0,2))} = \frac{1}{2}\cdot\int_0^2 F_\beta d\beta$$
$$F_{\text{Stability}(\beta\in(0,2))} = -\log\left[\frac{1}{2}\cdot\int_0^2 \left(F_\beta - F_{\text{Mean}(\beta\in(0,2))}\right)^2 d\beta\right]. \tag{24}$$

The leading period evaluates the maintaining of performance under all β values, as shown in the following equation:

$$\text{Leading Period}(c) = \frac{1}{2}\cdot\int_0^2 \delta_{\max}\left[F_\beta(c)\right]d\beta$$
$$\delta_{\max}\left[F_\beta(c)\right] = \begin{cases}1, & F_\beta(c) = \max_c\left(F_\beta(c)\right)\\ 0, & \text{otherwise}\end{cases}$$
$$c \in \begin{bmatrix}\text{INT, N-SS, ASM, CON,}\\ \text{ENT, Semirange, Saliency}\end{bmatrix}. \tag{25}$$

Another two indices, $\text{Recall}_{\text{FalseRate=1\%}}$ and $\text{FalseRate}_{\text{Recall=98\%}}$, are constructed based on the ROC curve. The former represents the recall value when the corresponding false rate in the ROC curve equals 1%, whereas the latter indicates the false rate when recall reaches 98%.

Table I illustrates the superiority of the proposed RSS for its excellent and robust performance in every testing scene with the best or the second-best score, followed by GLCM-based indicators, CON, and ENT. GLCM-CON shows a better best performance, whereas ENT shows a better average performance. This result also validates the effectiveness and robustness of these conventional texture indicators. Semivariogram-based semirange shows the capability of dealing with a complex and extremely complex background. Thus, semivariogram is still a powerful indicator and is commonly employed in SAR image processing. Spectral saliency as a global indicator is not suitable for detecting targets in complex backgrounds, but it performs well on weak targets' detection when the background is ideal. GLCM-ASM is not as powerful as CON and ENT, merely obtaining one second-optimal score, but it is still far better than the intensity data.

A custom score is constructed in (26) to show an overall evaluation of these seven indicators

$$\text{Custom Score} = \frac{\left(\text{OI} + \frac{\text{SOI}}{2}\right)\times(\text{Total Scenes} - \text{IS})}{\text{Total Items}} \tag{26}$$

where OI, SOI, and IS represent the number of optimal items, second optimal items, and invalid scenes, respectively; Total Scenes means the six testing scenes and equals 6; and Total Items equals 10, representing the ten evaluation items in each testing scene.

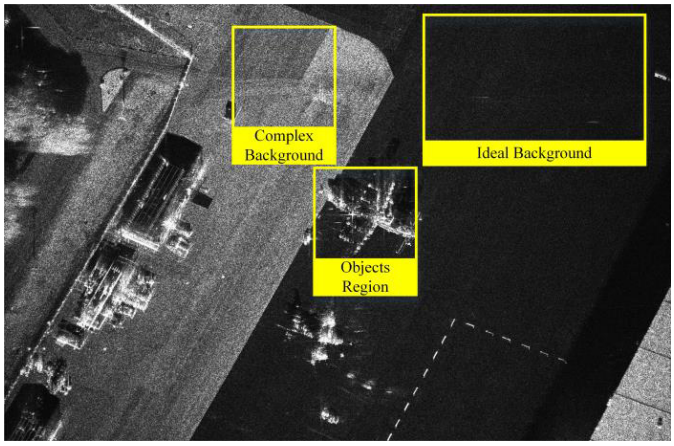


Fig. 17. MiniSAR image.

The detection experiments with different environmental parameters using samples in meter-level high-resolution SAR images show that the proposed RSS outperforms the canonical and commonly used indicators in both effectiveness and robustness aspects. Particularly, RSS shows good adaptability to targets with different scattering powers in backgrounds of various complexities. This highlight is essential and strongly demanded by applications of detecting targets in unknown regions with little prior field information.

B. Background Filtering Test Using VHR SAR Images

In very-high-resolution (VHR) SAR images of submeter resolutions, targets exhibit a different form and are far more complicated than those in meter-level SAR images. In VHR images, a target is a whole formed by parts with complex sizes and structures, occupying a region with a certain area. Meanwhile, targets or other scatters that are categorized into the background are presented more in detail in VHR images, increasing the background interference. Therefore, filtering the background interference is the main difficulty in VHR images, whereas target description is the main difficulty in HR images.

The VHR testing image, a Ku-band MiniSAR image provided by the Sandia National Laboratories with a resolution of 0.1 m, is shown in Fig. 17.

An airport is shown in Fig. 17, where the man-made targets of interest are airplanes and the corresponding facilities. The main interference and backgrounds are the runway and isolation zone noted by the ideal background and the complex background in Fig. 17. For target detection, both the runway and the isolation zone are considered interferences though they are totally different in scattering power and texture pattern.

The RSS map and comparative results are shown in Fig. 18. Pixel-based statistics with a certain threshold fail to evaluate the performance accurately due to the complex structure and distribution of targets. Hence, the Kullback–Leiber (KL) divergence that measures the difference between two distributions is employed as the criterion, as shown in the following equation [48]:

$$\text{KL}(R_1||R_2) = \int_{-\infty}^{+\infty} R_1(x)\log\frac{R_1(x)}{R_2(x)}dx \tag{27}$$

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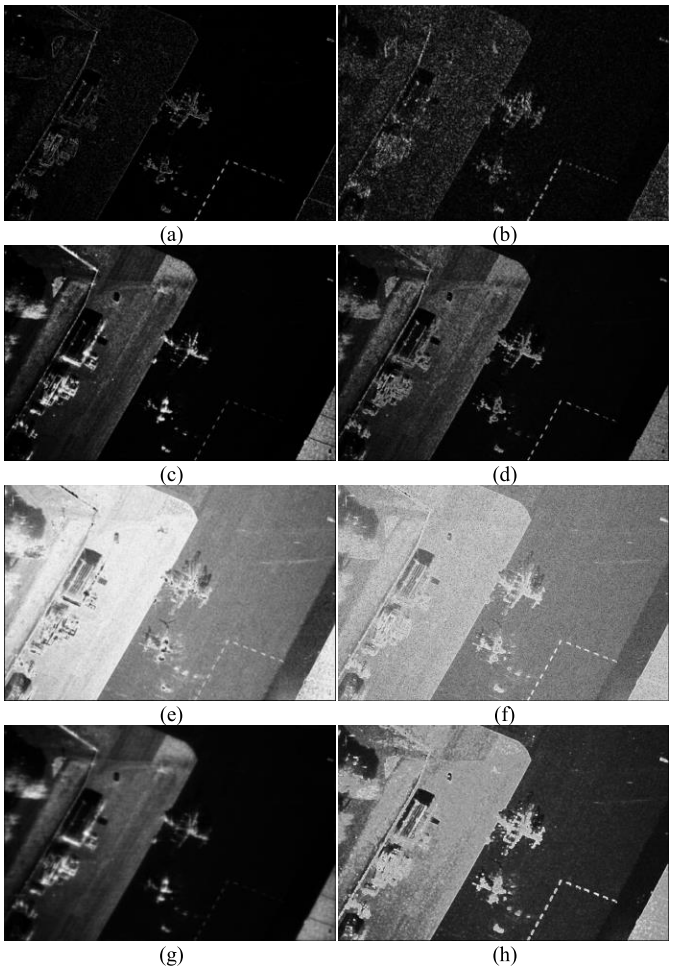


Fig. 18. Results of RSS and comparisons. (a) RSS map ($K = 0.96$). (b) Gabor filtered map. (c) ASM. (d) CON. (e) ENT. (f) Semirange. (g) Saliency. (h) Homogeneity.

where R_1 and R_2 stand for two distributions in research. A higher KL value between the target region and the background region means that it is easier to separate target pixels and background pixels. A low KL value between the complex and the ideal background regions represents that the indicator treats different backgrounds as one class though they are totally different in intensity and texture. Based on this principle, an overall KL-based score is designed as (28) to evaluate the general performance of background filtering

$$\text{Overall Score} = \frac{\text{KL}\,(O||I) \times \text{KL}\,(O||C)}{[\text{KL}\,(I||C)]^2} \tag{28}$$

where *O, I,* and *C* stand for targets region, ideal background, and complex background, respectively.

Experimental results are given in Fig. 19, where Fig. 19(a) shows the KL distance between the target region and the ideal background. This result reveals the simplicity of target detection under ideal conditions, where homogeneity and saliency take the leading place, followed by GLCM-based indicators. RSS indicator scores 197, which approaches ASM and contrast, illustrating that RSS acquires an acceptable performance in separating regions of targets and ideal background. The KL distance between the target region and the

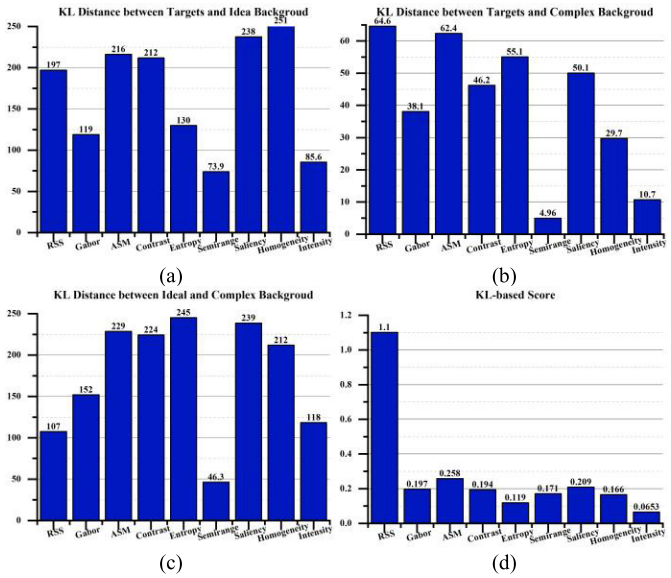


Fig. 19. (a) Difference between target regions and ideal background regions. (b) Difference between target regions and complex background regions. (c) Similarity between ideal and complex background regions. (d) Overall evaluation for target extraction and background filtering.

complex background is shown in Fig. 19(b), where RSS obtains the highest score of 64.6. The score is much lower than that in Fig. 19(a) due to the increasing background complexity. ASM takes second place, whereas contrast is surpassed by entropy. Homogeneity scores 29.7, showing that background complexity greatly impacts homogeneity. The KL distance between the ideal and the complex backgrounds is given in Fig. 19(c), where a lower score indicates a lower impact from the background complexity. Apart from semirange, RSS acquires the lowest score. This means that RSS treats the complex and the ideal backgrounds equally. Therefore, background regions of different complexity are more likely to be filtered simultaneously by RSS without additional processing. The overall scores are given in Fig. 19(d), and RSS outperforms others greatly.

The experimental results show the outstanding performance of RSS for filtering background and detecting man-made targets using VHR SAR images, whereas the increasing resolution and complexity degrade the capability of other indicators to some extent. Though RSS fails to acquire the best performance under every scenario, it shares the most robust performance, which is of high value in practical applications dealing with complex SAR images.

C. Man-Made Targets' Detection Using Various SAR Images

The MiniSAR image shown in Fig. 17 is employed to test the performance of the RSS-based targets' detection method. Based on the RSS map shown in Fig. 18(a), the SFFe is conducted in the next stage, and a simple noise filtering algorithm is employed in the postprocessing stage to show the performance of RSS-SFFe intuitively. The results are given in Fig. 20.

Owing to the outstanding performance of RSS for background filtering and targets' detection, the segmented and

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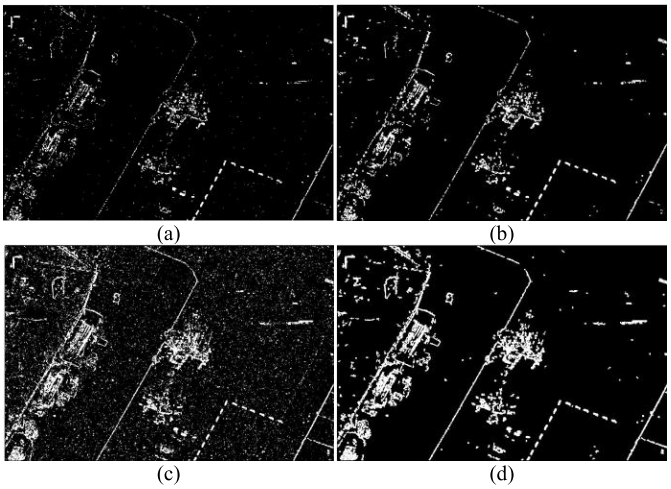


Fig. 20. Segmented and postprocessed RSS map with $e = 0.1\%$. (a) Segmented RSS map with $p = 5\%$. (b) Refined result of (a). (c) Segmented RSS map with $p = 10\%$. (d) Refined result of (c).

refined results in Fig. 20 are visually acceptable. Hence, the evaluation focuses on efficiency and parameter selection tolerance. The RSS map segmented by SFFe with sparsity rate $p = 5\%$ and error tolerance $e = 0.1\%$ is shown in Fig. 20(a), where ten iterations are consumed, the final threshold $T^* = 696.534$, the final sparsity rate $p^* = 2.59\%$, and the real error rate $e^* = 2.41\%$. The real error rate e^* is greater than the tolerance of 0.1%, indicating that there is no threshold other than 696.534 that can segment the RSS map within the error rate of 2.41%, and any other threshold will result in a greater error rate. A morphology-based algorithm is employed to filter isolated points in Fig. 20(a), and the processed result is given in Fig. 20(b). Another segmented result using $p = 10\%$ is shown in Fig. 20(c), where 12 iterations are consumed, the final sparsity rate p^* equals 10.04%, and the real error rate $e^* = 0.04\%$. Apparently, a higher sparsity rate parameter results in more extracted parts and noise points, like in Fig. 20(c). However, the same simple postprocessing algorithm that is employed to process Fig. 20(a) can perfectly filter the noise points and refine the result, as shown in Fig. 20(d). Both Fig. 20(b) and (d) shows that the man-made targets of interest and high value are well detected and extracted, meeting the requirements of man-made targets' detection and background filtering.

Comparing Fig. 20(b) and (d), the extracted parts are generally similar. The main differences are details in these regions, but these details are not necessary for target detection. This shows a high tolerance of parameter selection because different sparsity rates produce similar results, and $p = 10\%$ is recommended and is employed in the following experiments.

From the view of efficiency, about ten iterations are used to acquire results like Fig. 20(a) and (c). Compared with conventional segmentation algorithms, such as the adaptive OTSU method, the 8-bit quantization is required by OTSU, restricting the precision of the segmented result, and 256 iterations are consumed to obtain the optimal threshold. This means that SFFe is 25 times faster than the 8-bit OTSU method, and the efficiency and practical convenience of the RSS-based method is enhanced greatly by the employment of SFFe.

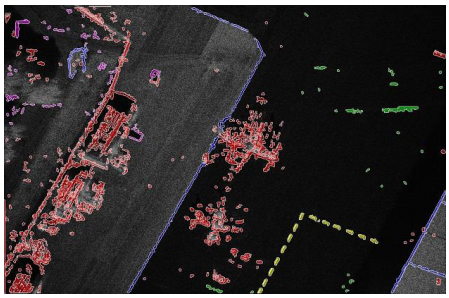
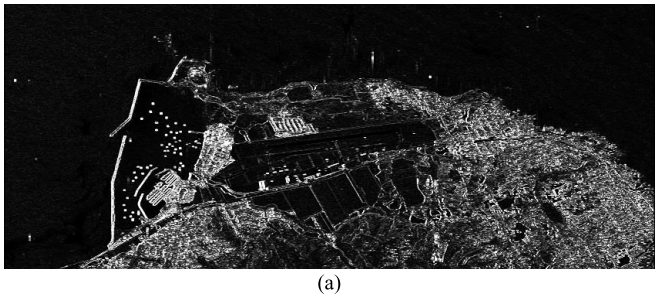


Fig. 21. Man-made targets' detection result using a 0.1-m-resolution VHR MiniSAR image.



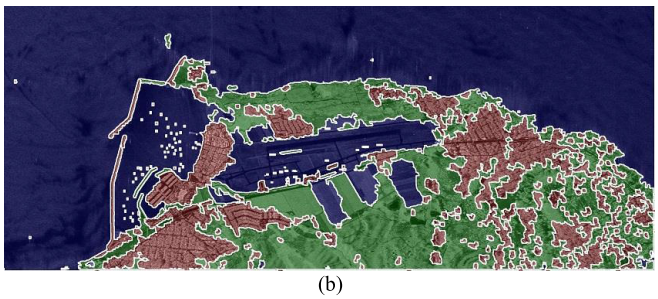


Fig. 22. RSS map and man-made targets' detection result using a 2-m-resolution HR UAVSAR image. (a) Enhanced display RSS map. (b) Final detection result.

A target recognition algorithm is employed to show the potential of the RSS-based method in real applications. The final target detection and classification result of Fig. 20(d) are given in Fig. 21. The red parts indicate the targets that are distinct from the background that they embed, including the airplanes, vehicles, ground facilities, and so on; the yellow regions are ground signs; the blue regions are edges between two different backgrounds; the green regions are noise but falsely detected as man-made targets by RSS-SFFe; and the purple regions are unknown parts.

The result shows that RSS-SFFe owns accepted performance on man-made targets' detection, where edges are the main false alarm; RSS-SFFe itself fails to recognize the certain classes of the detected targets, but these targets can be categorized with the help of a simple postprocessing algorithm, which is not the key research point in this article. The result also validates that RSS-SFFe is not vulnerable to noise due to both the powerful performance of RSS and the tolerance of SFFe.

The UAVSAR image in Fig. 4 is employed to test the performance of the RSS-based method on meter-level high-resolution SAR image, and the RSS map and the final result are given in Fig. 22. Because the dynamic range is so large that

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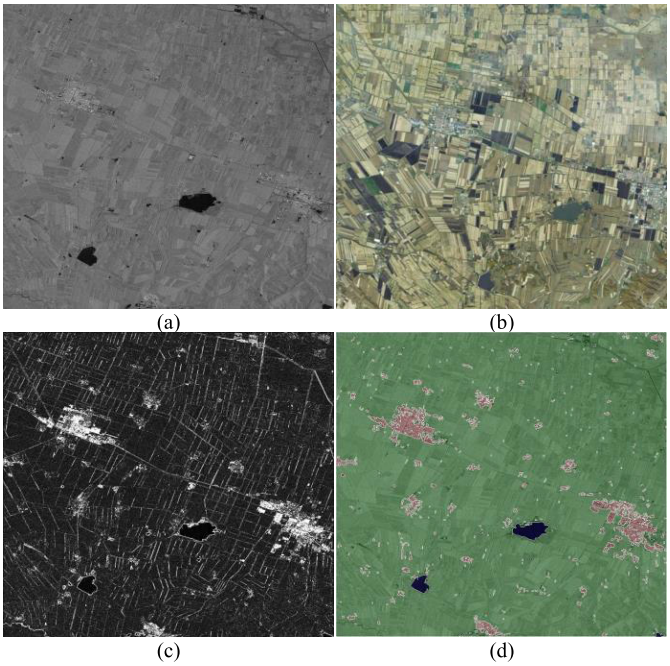


Fig. 23. RSS map and man-made targets' detection result using an 8-m-resolution HR GF-3 image. (a) Logarithmic filtered intensity image. (b) Optical image. (c) RSS map. (d) Final detection result.

the original RSS map, as shown in Fig. 7, is hard to display, an enhanced display is given in Fig. 22(a). In Fig. 22(b), the red regions indicate the detected man-made targets, the blue parts stand for the ideal background with low scattering power, and the green parts are common backgrounds. Restricted by resolution, it is difficult to divide the built-up regions into isolated buildings, so the detected results, apart from ships, are exhibited as regions. Though the employment of sophisticated postprocessing algorithms may acquire accurate contours and structures of building targets, the introduced algorithms fail to do the same work when image resolution changes, so the robustness is damaged seriously.

This result shows that most man-made targets, including buildings and urban areas in the complex mountainous background, isolated vehicles and facilities in the airport background, and ships in the ideal water background, are well detected, indicating that the proposed RSS-based method performs well using meter-level HR SAR images.

The C-band 8-m-resolution GaoFen-3 (GF-3) image of Youyi Farm in Heilongjiang Province in China is employed to test the performance on meter-level common SAR images, and the results are given in Fig. 23.

Fig. 23(a) and (b) shows the logarithm intensity image and the corresponding optical image of Youyi Farm, respectively; (c) is the RSS map, and (d) is the final result where the red regions indicate man-made targets, the blue parts are water regions, and the green is the farmland background. The man-made target in research, including urban and human active areas, isolated facilities scattered in the farmland, and unknown singular targets, are all well detected and marked with accepted accuracy by the RSS-based method. This result illustrates that the RSS-based method is still effective for 10-m-level SAR images.

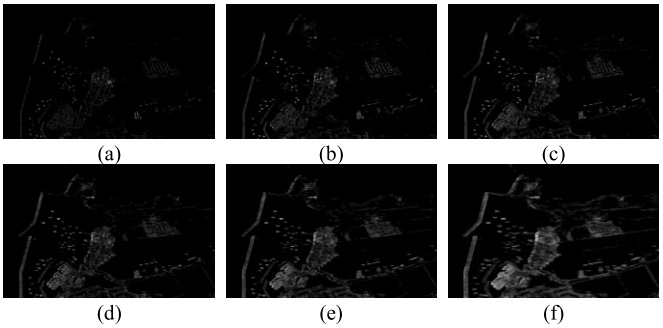


Fig. 24. Impact of d on the spatial resolution. (a) $d = 3$. (b) $d = 5$. (c) $d = 7$. (d) $d = 9$. (e) $d = 11$. (f) $d = 15$.

The three detection experiments on submeter VHR SAR image (MiniSAR), 2-m HR SAR image (UAVSAR), and 8-m SAR image (GF-3) show that the proposed RSS and the corresponding detection method can detect man-made targets in different backgrounds with an accepted overall performance. Meanwhile, the RSS-based method maintains a good performance on a wide range of resolutions and bands, which cover most SAR images. Though there is still a gap between the proposed method and the current target detection methods in detail preserving, the proposed method is robust to deal with various SAR images and shows great practical potential for SAR applications.

D. Parameter Discussion and Computational Complexity

There are only three tunable parameters in the proposed method: the scale parameter d, the truncation rate K, and the estimated sparsity p. The less tunable parameters mean a more stable method in practice, but the adaptivity and generalization are also limited. The experiments above validate that the proposed three-parameter method achieves acceptable overall performance on most SAR images. Then, in this section, the parameter impact, the selection strategy, and recommended values are discussed to show the high tolerance of parameter selection and the excellent robustness and generalization of our method.

The scale parameter d refers to the radius of the sampling window, and the window size equals $(2d + 1)$. Thus, d is related to the resolutions of both the space and the spectrum, and the selection is a tradeoff between them. Fig. 24 shows the impact of d on spatial resolution, where a larger d leads to a lower resolution and a more blurred result. In this view, a smaller d is preferred to reserve the spatial details.

However, a too small d deteriorates the spectral resolution and the performance of the RSS. The same experimental scheme and six scenes in Section V-A are employed, and the ROC curves are used to show the impact of d on RSS performance, as shown in Fig. 25. Fig. 25(a), (b), (d), and (e) shows that a larger d acquires better performance if the background is not extremely complex, where the performance of $d = 5$ is much better than $d = 3$, but the performance increases little when d is over 7. Fig. 25(c) and (f) reveals that $d = 5$ and $d = 7$ are better than others and, for the spatial details, are well preserved. These phenomena are in accordance with the

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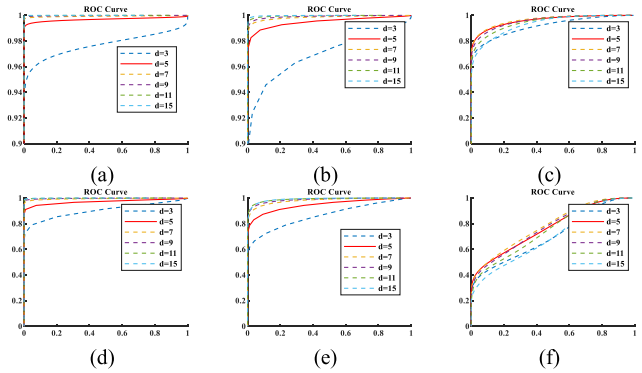


Fig. 25. Impact of d on the RSS performance. (a)−(f) scene 1–scene 6.

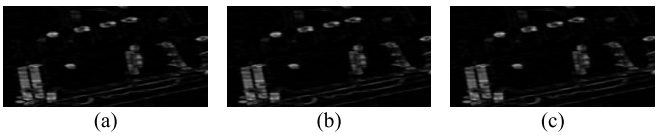


Fig. 26. Impact of K on the UAVSAR image. (a) $K = 0.92$. (b) $K = 0.96$. (c) $K = 1$.

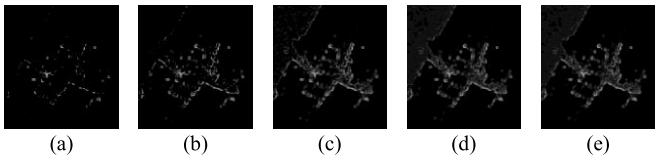


Fig. 27. Impact of K on the MiniSAR image. (a) $K = 0.92$. (b) $K = 0.94$. (c) $K = 0.96$. (d) $K = 0.98$. (e) $K = 1$.

previous work on the tradeoff between spatial and spectral resolutions [31]. Thus, [4, 9] is a recommended range where $d = 7$ may have the best overall performance. However, the scale parameter d is also related to the time consumption. Hence, $d = 5$ is employed in our work for a tradeoff between effectiveness and efficiency.

The parameter K in (15) is to enhance the capability of complex homogeneous background filtering. In practice, the impacts of K are different on images of different resolutions. For meter-level SAR images, there is no significant difference among different K's, as shown in Fig. 26. In contrast, the impacts of K are more obvious for submeter VHR SAR images, as shown in Fig. 27, where the background parts above the aircraft are well filtered, but the details of targets are also lost. That is because the background parts are more complicated when the resolution is up to the submeter level. Hence, the overall data chaos increase, leading to high spectral entropy. This phenomenon shows the significant differences between images of different resolutions, partially illustrating the difficulty of formulating a method simultaneously suitable for images of various resolutions.

The selection of K is also related to d. When d increases, the sampling window is larger, and the information is more chaotic, then the overall entropy increases. Thus, the same K may filter more details when a larger d is employed, as shown in Fig. 28.

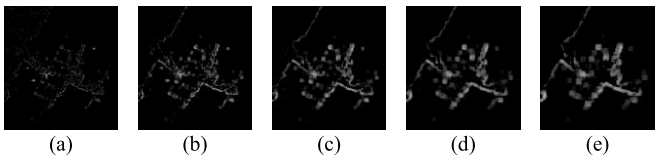


Fig. 28. Relationship between K and d. (a) $d = 3$. (b) $d = 5$. (c) $d = 7$. (d) $d = 9$. (e) $d = 11$.

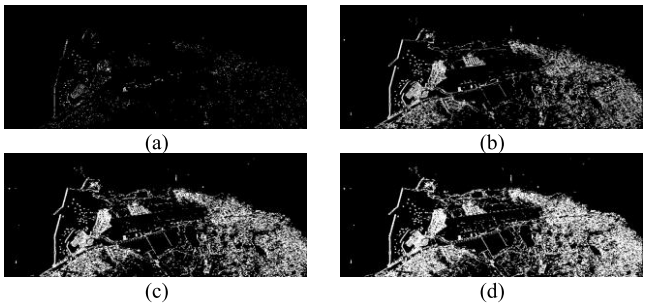


Fig. 29. Impact of p on the UAVSAR image. (a) $p = 1\%$. (b) $p = 10\%$. (c) $p = 15\%$. (d) $p = 25\%$.

According to the mechanism, theoretical analysis, and experimental results, 0.96 is recommended when $d = 5$ because the value is simultaneously suitable for the meter- and submeter-level images.

The variable p in (17) is the preset sparsity parameter, and the impact of p is shown in Fig. 29. When $p = 1\%$, most land targets are missed. When p is over 10%, the main bodies of man-made targets are well detected. There is no significant change when p changes from 15% to 25% other than more details in target regions and more noise points. Similar results of the MiniSAR image can also be seen in Fig. 20. The results illustrate that p can be selected in a wide range to obtain similar results, validating the high parameter tolerance of the proposed SFFe method. According to the experiments and practical experience, p can be selected within [5%, 20%], and the recommended value is 10%.

The computational complexity is not the highlight of the proposed method because of the spectrum-based mechanism. The main time-consumed procedure is FFT-based spectrum extraction, so the computational complexity of one calculation unit is $O(R^2\log_2(R))$, where R is the unit size and $R = (2d + 1)$. The computational complexity of a space-based method is $O(R^2)$, such as the convolution-based Gabor filter and the second-order-statistic-based spatial homogeneity. The $O(R^2)$ complexity is generally considered a low computational burden. Theoretically, the complexity ratio between our method and the $O(R^2)$ method is $O(R^2\ \log_2\ (R))/O(R^2)$, and the ratio is $\log_2(R)$ if the coefficient and remainders are neglected. For example, if $d = 3 (R = 7)$, the ratio approximately equals $\log_2\ (7) \approx 2.8$.

The intuitive time consumption and comparison between our method and an $O(R^2)$ method are given in Fig. 30, where the colored bars show the time consumption of our method; the gray bars show the time consumption of the $O(R^2)$ method; the left-horizontal axis stands for parameter d; the right-horizontal axis shows the sizes of testing images;

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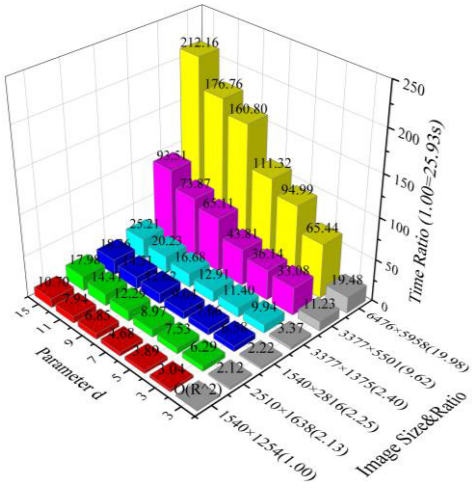


Fig. 30. Computational complexity and time-consuming analysis.

the values in brackets denote the size ratio with the basis of 1540×1254; and the vertical axis stands for the time consumption ratio with the basis of 25.93 s.

The results in Fig. 30 validate the theoretical analysis, as the time ratio between our method and the $O(R^2)$ is approximately 3. In the size dimension, the time consumption linearly increases, and the time ratio is in accordance with the size ratio. In the parameter dimension, the time consumption increases with d. There are sharp increases when d changes from 3 to 5 and 7 to 9. That is because of the mechanism of base-2 FFT, where the eight-point FFT changes to the 16-point and 32-point when the size changes from 7 to 11 and 15 to 19. Generally, the computational complexity is not our highlight, but the time consumption is so stable that its expectation can be estimated by theoretical analysis in advance. This means that the time consumption is controllable and predictable.

According to the parameter discussion and computational complexity analysis, it is validated that our method can obtain good overall performance on various SAR images with a high-tolerance parameter selection and predictable time consumption. Notably, all the experimental results are based on one set of parameters, $d = 5$, $K = 0.96$, and $p = 10\%$, showing great robustness, stability, and generalization. Such properties are of great importance in practical applications.

VI. Conclusion

This article analyzes the scattering of man-made targets and natural backgrounds in SAR images and assumes that the noise characteristics remain relatively stable and invariant when resolution, pose, and background change. Then, taking the manually generated noise with known properties as a standard, a similarity between the observing data and ideal noise is formulated based on the spectral stationarities in two dimensions. The RSS is proposed based on this similarity to serve as the basis for man-made targets' detection, where a higher RSS value indicates a greater distance to noise and a higher likelihood to be a part of man-made targets. A fast and refined segmentation algorithm named SFFe is proposed to process the RSS data efficiently. SFFe can deal with the data without format restriction and solves the accuracy loss caused by histogram estimation. A detection framework is established combining RSS and SFFe.

The UAVSAR, MiniSAR, and GF-3 SAR images of resolutions ranging from 0.1 to 8 m, covering the C-, L-, and Ku-bands, are employed to test the performance of RSS and the corresponding detection framework. Six common practical scenes with different detection difficulties are constructed based on the combination of man-made targets with strong or weak scattering power and backgrounds with ideal, complex, and extremely complex contents. The quantitative results illustrate that the RSS indicator detects man-made targets well in all six testing scenes and outperforms other canonical indicators under various criteria, validating its detection performance. The experiment of background suppressing validates that the RSS indicator suppresses and filters backgrounds with both ideal and complex contents, acquiring a better KL-based score than other indicators. The detection experiments validate that the RSS-based detection framework works well on different images of various image parameters and content. All these comprehensive experiments show the effectiveness, reliability, robustness, and generality of the proposed indicator and the detection framework. The highlight is that the framework maintains good performance on a wide span of resolutions covering nearly all commonly used resolutions in SAR images. Notably, all the experimental results are based on one set of parameters, $d = 5$, $K = 0.96$, and $p = 10\%$, showing great robustness, stability, and generalization. Such properties are of great importance in practical applications. With the assistance of other sophisticated and powerful preprocessing and postprocessing methods, the indicator and framework will acquire more powerful capability and developing potential.

In future research, the theoretical model will be analyzed more deeply to strengthen the capability for image details describing and reduce the reliance on preprocessing and postprocessing. Reducing the computational burden and increasing efficiency are also taken into consideration in the following research.

Appendix

According to (3)–(7), the input $\mathbf{I}$ can be represented as the summation of object scattering $\mathbf{O}$ and noise $\boldsymbol{N}$, where the noise also can be divided into a constant noise map $\bar{\boldsymbol{N}}$ and the remainder $\boldsymbol{N}^*$, as shown in (A.1), and the statistical restrictions are listed in (A.2), where c_1 and c_2 denote constant values

$$\left.\begin{aligned}\mathbf{I} &= \mathbf{O} + \boldsymbol{N} \\ \boldsymbol{N} &= \bar{N} + \boldsymbol{N}^*\end{aligned}\right\} \Rightarrow \mathbf{I} = \mathbf{O} + \bar{N} + \boldsymbol{N}^* \tag{A.1}$$

$$\begin{aligned}&\mathbf{O} \equiv E(\mathbf{O}) \equiv c_1, \bar{N} \equiv E(\boldsymbol{N}) = c_2, E\left(\boldsymbol{N}^*\right) = 0 \\ &E(\boldsymbol{N}) = E\left(\bar{N} + \boldsymbol{N}^*\right), E(\mathbf{I}) = E(\mathbf{O} + \boldsymbol{N}) = c_1 + c_2.\end{aligned} \tag{A.2}$$

According to the properties of Fourier transform, the spectrum of input $\mathbf{I}$ can be written as (A.3), where $\mathbf{O}$ and $\bar{N}$ only contribute to the dc component in the whole spectrum

$$\mathbf{F}_\mathbf{I} = \mathbf{F}_\mathbf{O} + \mathbf{F}_{\boldsymbol{N}} = \mathbf{F}_\mathbf{O} + (\mathbf{F}_{\bar{N}} + \mathbf{F}_{\boldsymbol{N}^*}) = (\mathbf{F}_\mathbf{O} + \mathbf{F}_{\bar{N}}) + \mathbf{F}_{\boldsymbol{N}^*}. \tag{A.3}$$

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We vectorize the 2-D spectrum matrix into a 1-D vector for a simplified deduction, and the specific forms of each spectrum component are shown in (A.4), where the subscripts $1 \sim L$ mathematically stand for the dimension of each vector and physically denote the frequency component. The first element in each vector stands for the dc component. The summated dc component, F_{dc}, is generated by the mean power of the input $\mathbf{I}$

$$\left.\begin{aligned} \mathbf{F_I} &= (f_1, f_2, \ldots, f_L)_{1\times L} \\ \mathbf{F_O} &= (f_{\mathbf{O}}, 0, \ldots, 0)_{1\times L} \\ \mathbf{F}_{N^*} &= (0, f_2, \ldots, f_L)_{1\times L} \\ \mathbf{F}_{\bar{N}} &= (f_{\bar{N}}, 0, \ldots, 0)_{1\times L} \end{aligned}\right\} \Rightarrow \begin{aligned} F_{\mathrm{dc}} &\triangleq f_1 = f_{\mathbf{O}} + f_{\bar{N}} \\ \mathbf{F}_{\mathrm{dc}} &\triangleq \mathbf{F_O} + \mathbf{F}_{\bar{N}} \end{aligned}. \tag{A.4}$$

Then, the statistics of the spectrum of input data are shown in (A.5), and the corresponding deductions are given in (A.6)–(A.8)

$$\begin{aligned} &\sigma_{\mathbf{F}}^2 \\ &\quad = \frac{1}{L-1}\sum_{i=1}^{L}(f_i - E_{\mathbf{F}})^2 \\ &E_{\mathbf{F}} = E(\mathbf{F_I}) = \frac{1}{L}\sum_{i=1}^{L} f_i = \frac{F_{\mathrm{dc}}}{L} + \frac{1}{L}\sum_{i=1, f_1=0}^{L} f_i \\ &\quad = \frac{F_{\mathrm{dc}}}{L} + E(\mathbf{F}_{N^*}) = \frac{F_{\mathrm{dc}}}{L} + E_N \end{aligned} \tag{A.5}$$

$$\begin{aligned} &(L-1)\,\sigma_{\mathbf{F}}^2 \\ &\quad = \sum_{i=1}^{L}(f_i - E_{\mathbf{F}})^2 \\ &\quad = \sum_{i=1}^{L}\left(f_i - E_N - \frac{F_{\mathrm{dc}}}{L}\right)^2 = \sum_{i=1}^{L}\left((f_i - E_N) - \frac{F_{\mathrm{dc}}}{L}\right)^2 \\ &\quad = \sum_{i=1}^{L}(f_i - E_N)^2 + \sum_{i=1}^{L}\left(\frac{F_{\mathrm{dc}}}{L}\right)^2 - \frac{2F_{\mathrm{dc}}}{L}\sum_{i=1}^{L}(f_i - E_N) \\ &\quad = (1) + \frac{F_{\mathrm{dc}}^2}{L} + (2) \end{aligned} \tag{A.6}$$

$$\begin{aligned} &(1) \\ &\quad = \sum_{i=1}^{L}(f_i - E_N)^2 = (F_{\mathrm{dc}} - E_N)^2 + \sum_{i=2}^{L}(f_i - E_N)^2 \\ &\quad = F_{\mathrm{dc}}^2 - 2F_{\mathrm{dc}}E_N + E_N^2 + \sum_{i=2}^{L}(f_i - E_N)^2 \\ &\quad = F_{\mathrm{dc}}^2 - 2F_{\mathrm{dc}}E_N + \left(0 - E_N^2\right) + \sum_{i=2}^{L}(f_i - E_N)^2 \\ &\quad = F_{\mathrm{dc}}^2 - 2F_{\mathrm{dc}}E_N + \sum_{i=1, f_1=0}^{L}(f_i - E_N)^2 \\ &\quad = (L-1)\,\sigma^2(\mathbf{F}_{N^*}) + F_{\mathrm{dc}}^2 - 2F_{\mathrm{dc}}E_N \end{aligned} \tag{A.7}$$

$$\begin{aligned} &(2) \\ &\quad = -\frac{2F_{\mathrm{dc}}}{L}\sum_{i=1}^{L}(f_i - E_N) = -\frac{2F_{\mathrm{dc}}}{L}\left(\sum_{i=1}^{L} f_i - \sum_{i=1}^{L} E_N\right) \\ &\quad = -\frac{2F_{\mathrm{dc}}}{L}(L\cdot E_{\mathbf{F}} - L\cdot E_N) = -2F_{\mathrm{dc}}(E_{\mathbf{F}} - E_N) \\ &\quad = -2F_{\mathrm{dc}}\cdot\frac{F_{\mathrm{dc}}}{L} \\ &\quad = -2\frac{F_{\mathrm{dc}}^2}{L}. \end{aligned} \tag{A.8}$$

Finally, let the dimension L equal the size of the sliding window $(2d+1)^2$; then, (9) is derived by (A.9). Thus, if $F_{\mathrm{dc}} = 0$, the deviation of the input spectrum can be totally represented by the noise spectrum

$$\begin{aligned} \sigma_{\mathbf{F}}^2 &= \frac{1}{L-1}\left((1) + \frac{F_{\mathrm{dc}}^2}{L} + (2)\right) \\ &= \frac{\left((L-1)\sigma^2(\mathbf{F}_{N^*}) + \frac{L-1}{L}F_{\mathrm{dc}}^2 - 2F_{\mathrm{dc}}E_N\right)}{L-1} \\ &= \sigma^2(\mathbf{F}_{N^*}) - \frac{2F_{\mathrm{dc}}E_N}{L-1} + \frac{F_{\mathrm{dc}}^2}{L}|_{L=(2d+1)^2} \\ &= \sigma^2(\mathbf{F}_{N^*}) - \frac{F_{\mathrm{dc}}\cdot E(\mathbf{F}_{N^*})}{2d(d+1)} + \frac{F_{\mathrm{dc}}^2}{(2d+1)^2}. \end{aligned} \tag{A.9}$$

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